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WORK EXPERIENCE

Undisclosed Hedge Fund **Jun 2024 – Oct 2024**
Data Operator Intern

- **Data Collection & Cleaning**
  - Collected raw data from multiple online and internal sources.
  - Cleaned and validated datasets to ensure consistency and usability.
  - Maintained and updated internal data inventories regularly.
- **Data Extraction & Annotation**
  - Extracted relevant data points based on senior analysts' guidance.
  - Annotated datasets for downstream processing and analysis.
  - Organized extracted data into structured formats for modeling use.

DLSU Center for Complexity and Emerging Technologies (COMET) **Jul 2024 – Aug 2024**

- **Geospatial Data Annotator**
  - Labeled satellite images in CVAT to identify informal settlements and redefine boundaries in Makati and Taguig. Created square grid overlays on city polygons to assist in spatial analysis and urban planning.

EDUCATION

DE LA SALLE UNIVERSITY – MANILA **2021 – Present**
Bachelor of Science in Computer Science Major in Software Technology

- Honorable Mention Standing; CGPA: 3.389/4.0
- Outstanding Thesis Award - Silver

LORMA SENIOR HIGH SCHOOL **2019 - 2021**
STEM Strand

- With High Honors; CGPA: 94.35

LORMA COLLEGES SPECIAL SCIENCE HIGH SCHOOL **2017 - 2019**

- With High Honors; CGPA: 93.5

REGENT SECONDARY SCHOOL – SINGAPORE **2016 - 2017**

- Ranked 8 in Express Stream Batch for Mid-Year Examination

MARSILING PRIMARY SCHOOL – SINGAPORE **2010 - 2016**

- PSLE Score: 192; Ranked 92nd in Whole Secondary Batch

PAST PROJECTS

CamoFarmer: A Lightweight Computer Vision Model for Camouflaged Crop Detection

- A research project focused on identifying camouflaged crops in natural farm environments using efficient deep learning models. The project explores architectures such as SSD, YOLOv8, and RT-DETR while applying model compression techniques including backbone replacement, pruning, and knowledge distillation to improve detection accuracy with lower computational cost. Models are trained on TomatOD, CCROP, and CamoCrops datasets for deployment on resource constrained platforms like drones and edge devices.
  - Contributed to the development of a lightweight object detection model for camouflaged crops, introduced the research topic, and contributed to key architectural decisions.
  - Preprocessed the TomatOD dataset for model training, handling image-label alignment, normalization, and annotation formatting.
  - Led all experiments and development involving the SSD model, including architecture selection, compression strategies, and performance benchmarking.

ETL and OLAP Dashboard Development for SeriousMD Data using Apache NiFi and Tableau

- A data engineering project focused on building an OLAP dashboard by processing the SeriousMD dataset through Apache NiFi for ETL. The data was structured into a star schema warehouse, enabling efficient analytics and dashboard creation in Tableau. Performance improvements were validated through functional testing and optimized SQL queries using foreign key indexing.
  - Processed and cleaned large-scale datasets containing millions of records.
  - Designed and executed ETL workflows in Apache NiFi to populate a star schema warehouse.
  - Built interactive Tableau dashboards for data visualization and insights.
  - Conducted functional and performance testing to validate query efficiency.

Classifying Celestial Objects Using Machine Learning Models

- A machine learning project that utilizes the Stars Dataset to classify celestial objects based on their physical attributes. The study evaluates model performance by training and testing various classifiers to identify the most effective approach for accurate classification.
  - Cleaned and prepared the dataset for modeling by handling missing values and formatting inconsistencies.
  - Conducted exploratory data analysis to identify key patterns and feature distributions.
  - Trained and evaluated a logistic regression model to classify celestial objects based on selected features.

Automated Fruit Maturity Classification with Deep Learning

- A Python-based research project that uses Convolutional Neural Networks to classify the maturity stage of Cavendish Bananas using image data.
  - Built and trained a CNN model in Jupyter Notebook to detect banana maturity stages from labeled images.
  - Used Python libraries such as TensorFlow and Keras for model development and evaluation.
  - Visualized model predictions and performance metrics to validate classification accuracy.

SKILLS

- **Programming Languages:** Python, SQL, Java, C++, C
- **Web & Mobile Development:** HTML, CSS, JavaScript, Java, Firebase
- **Computer Vision & Machine Learning:** Object Detection, Image Classification, Image Segmentation, Model Training & Fine-Tuning, Dataset Preprocessing, Model Evaluation
- **Frameworks & Libraries:** PyTorch, YOLOv8, OpenCV, NumPy, Pandas
- **Data Science:** Statistical Analysis, Data Mining, Data Visualization, Machine Learning
- **Tools:** MS Workspace, Google Workspace, Git/GitHub, Figma, Tableau, Flourish, MySQL, Jupyter Notebook (with Pandas), Apache Nifi (ETL)